

IA Mathematics for Natural Sciences: Examples Sheet 0: Basic Skills

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Making friends

1. Speak to your supervision partner, and, if you don't know already, find out: (a) where they are from; (b) their favourite food; (c) what they like to do to relax; (d) what part of the maths course they are most excited about this year. [If you have the misfortune of being a singleton, ask someone else in your year instead!]

Basic logic

2. Explain the meaning of the logical symbols $\Rightarrow$, $\Leftarrow$ and $\Leftrightarrow$. [If you haven't seen them before, look them up online! In general, you should feel free to look up terms you don't understand on an examples sheet.] Decide which of the following are true:

$$(a) x^2 \leq 1 \Rightarrow x \leq 1, \quad (b) x^2 \leq 1 \Leftarrow x \leq 1, \quad (c) x^2 \leq 1 \Leftrightarrow x \leq 1.$$

Similarly, decide which of the following are true:

$$(d) |x| = 1 \text{ if } x = 1, \quad (e) |x| = 1 \text{ only if } x = 1, \quad (f) |x| = 1 \text{ if and only if } x = 1.$$

3. What is the error in the following argument?

'Suppose we wish to solve the equation $x - 1 = 2$. We begin by squaring both sides, to obtain $(x - 1)^2 = 4$. Expanding the left hand side, we have:

$$x^2 - 2x + 1 = 4 \quad \Leftrightarrow \quad x^2 - 2x - 3 = 0 \quad \Leftrightarrow \quad (x + 1)(x - 3) = 0.$$

Since one of these factors must be zero, it follows that there are two solutions to the equation, $x = -1$ or $x = 3$.'

4. What is meant by *proof by contradiction*? Prove by contradiction that $\sqrt{2}$ is irrational.

Sets and functions

5. State what is meant by the sets $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$. Decide which of the following statements are true:

$$(a) \pi \in \mathbb{Q}, \quad (b) 3 \notin \mathbb{R}, \quad (c) \mathbb{Z} \subseteq \mathbb{Q}, \quad (d) \mathbb{Q} \supset \mathbb{C}.$$

6. State what is meant by the sets $[a, b]$, (a, b) , and $(a, b]$, where a, b are real numbers. Decide which of the following statements are true:

$$(a) 1 \in [0, 1), \quad (b) 3 \notin (3, 4), \quad (c) [2, 3] \subset (2, 5], \quad (d) (-1, 0) \subset [-1, 0].$$

7. A function is a mapping from a set to another set. We write $f : A \rightarrow B$ to denote the function (or just f when the sets are implied), and we write $f(x)$ for the value of the function at the point $x \in A$. Which of the following define functions, and why?

(a) $f : [0, \infty) \rightarrow \mathbb{R}$ given by $f(x) = \sqrt{x}$;

(b) $f : \mathbb{Z} \rightarrow \mathbb{Z}$ given by $f(x) = \frac{1}{2}x$;

(c) $f : \mathbb{R} \rightarrow \mathbb{R}$ where $f(x)$ is implicitly defined by $2^{f(x)} = x$;

(d) $f : [0, \infty) \rightarrow \mathbb{R}$ where $f(x)$ is implicitly defined by $f(x)^2 = x$.

Sequences and series

8. Prove the formula:

$$\sum_{k=1}^n k = \frac{1}{2}n(n+1)$$

by (a) induction; (b) considering the sum of $(k+1)^2 - k^2$ from $k = 1$ to $k = n$. Is it more useful to prove a result by induction, or by a direct argument? Why?

9. What is meant by an *arithmetic sequence* with first term a and common difference d ? Prove that the sum of the first n terms of an arithmetic sequence is

$$S_n = \frac{1}{2}n(2a + (n-1)d).$$

Hence, find the sum of the series 2, 5, 8, 11, ..., 32.

10. What is meant by a *geometric sequence* with first term a and common ratio r ? Prove that if $r \neq 1$, the sum of the first n terms of a geometric sequence is:

$$S_n = \frac{a(1-r^n)}{1-r}.$$

What happens if $r = 1$? What is the behaviour of the sum in the limit as $n \rightarrow \infty$? Hence, find the sum of the infinite series $2, 2/3, 2/9, 2/27, \dots$

11. Using the formula for the sum of an infinite geometric series, find a formula for the sum of the infinite series:

$$\sum_{k=1}^{\infty} kr^k,$$

where $|r| < 1$. Hence, determine $2/3 + 2(2/3)^2 + 3(2/3)^3 + 4(2/3)^4 + \dots$.

Trigonometric functions and their identities

12. Define the trigonometric functions $\sin(x), \cos(x), \tan(x)$ using the sides of a right-angled triangle. Define also the reciprocal trigonometric functions $\operatorname{cosec}(x), \sec(x), \cot(x)$. Hence, prove each of the following trigonometric identities:

(a) *The Pythagorean identities:* $\sin^2(x) + \cos^2(x) = 1, \tan^2(x) + 1 = \sec^2(x), \cot^2(x) + 1 = \operatorname{cosec}^2(x)$.

(b) *The cofunction identities:* $\sin(\frac{\pi}{2} - x) = \cos(x), \cos(\frac{\pi}{2} - x) = \sin(x), \tan(\frac{\pi}{2} - x) = \cot(x)$.

(c) *The compound angle formulae:*

$$\sin(x \pm y) = \sin(x)\cos(y) \pm \sin(y)\cos(x), \quad \cos(x \pm y) = \cos(x)\cos(y) \mp \sin(x)\sin(y), \quad \tan(x \pm y) = \frac{\tan(x) \pm \tan(y)}{1 \mp \tan(x)\tan(y)}.$$

(d) *The double angle formulae:*

$$\sin(2x) = 2\sin(x)\cos(x), \quad \cos(2x) = \cos^2(x) - \sin^2(x) = 2\cos^2(x) - 1 = 1 - 2\sin^2(x).$$

(e) *The power reduction formulae:* $\sin^2(x) = \frac{1}{2}(1 - \cos(2x)), \cos^2(x) = \frac{1}{2}(1 + \cos(2x))$.

Learn all of these identities off by heart, and get your supervision partner to test you on them.

Exams and coursework

13. (a) Look up the format of the first-year maths exams. How much is each exam worth? How long does each exam last? How many questions do you have to answer, and how should you split your time in the exams? Are you allowed a calculator? Is there a formula book?
- (b) Look up the first-year maths coursework. How much is it worth?